# The genus-1 seven faces: elliptic collision residues and the KZB connection

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**Abstract**

At genus 0, the collision residue of a chirally Koszul chiral algebra admits seven equivalent presentations: as a twisting morphism, a line-operator $r$-matrix, a $\lambda$-bracket, commuting sphere Hamiltonians, the Drinfeld $r$-matrix, the Sklyanin bracket, and the Gaudin Hamiltonians. We prove the genus-1 extension: on an elliptic curve $E_\tau$, the genus-1 collision residue $r^{(1)}_{\mathcal{A}}(z,\tau)$ is the elliptic regularization of the genus-0 residue, obtained by replacing rational propagator kernels with Weierstrass functions. For affine Kac–Moody algebras $\mathcal{A} = \widehat{\mathfrak{g}}_k$, the seven faces specialize to the Knizhnik–Zamolodchikov–Bernard connection, the Belavin–Drinfeld elliptic $r$-matrix, and the elliptic Gaudin Hamiltonians. The $B$-cycle monodromy of the collision residue recovers the quantum group parameter $q = \exp(2\pi i/(k+h^\vee))$. For class $M$ algebras (Virasoro, $\mathcal{W}_N$), the collision residue produces higher-order elliptic differential operators involving $\wp, \wp', \ldots, \wp^{(2N-3)}$ that have no counterpart in the existing integrable systems literature.

## 1 Introduction

The genus-0 seven-face theorem [8] identifies the collision residue $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ of a chirally Koszul chiral algebra $\mathcal{A}$ with seven classical objects: a bar-cobar twisting morphism, a line-operator $r$-matrix in the Dimofte–Niu–Py framework [4], a classical $\lambda$-bracket in the Khan–Zeng 3d sigma-model [6], the Gaiotto–Zeng commuting sphere Hamiltonians, the Drinfeld $r$-matrix, the Sklyanin Poisson bracket, and the Gaudin Hamiltonians. The seven presentations agree as elements of $\mathrm{End}_{\mathcal{A}}(2)[\![z^{-1}]\!]$; their mutual identification is controlled by the collision depth $k_{\max}$ and the shadow class $G/L/C/M$.

The collision residue $r_{\mathcal{A}}(z)$ is the degree-2 component of the $E_1$ Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$ in the ordered convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$, where $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the cofree tensor coalgebra with deconcatenation coproduct. Its $\Sigma_n$-coinvariant projection is the scalar shadow $\kappa$: $\mathrm{av}(r(z))$ in abelian and scalar families, and $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody; the spectral parameter that $\kappa$ discards is precisely the data the seven faces read. At genus 1, the spectral parameter acquires modular dependence: $r^{(1)}_{\mathcal{A}}(z,\tau)$ depends on the elliptic modulus $\tau$, and the $B$-cycle monodromy of the ordered collision residue encodes the quantum group parameter.

This paper extends the identification to genus 1. The elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ replaces the Riemann sphere, the rational propagator $1/z$ is replaced by the Weierstrass zeta function $\zeta_\tau(z)$, and each rational function $1/z^{n+1}$ in the collision expansion is replaced by its canonical elliptic counterpart. The resulting genus-1 collision residue $r^{(1)}_{\mathcal{A}}(z,\tau)$ admits seven equivalent presentations that degenerate to the genus-0 faces as $\tau \to i\infty$.

The main results are:

(1) The *elliptic regularization theorem* (Theorem 2.4): the genus-1 collision residue is

$$r_{\mathcal{A}}^{(1)}(z,\tau) \;=\; c_0\,\zeta_\tau(z) \;+\; \sum_{n\geq 1} \frac{(-1)^{n-1}}{n!}\, c_n \cdot \wp^{(n-1)}(z,\tau),$$

where $c_n = a_{(n)}b$ are the OPE-mode collision coefficients.

(2) The identification of the $dz$-component of the *KZB connection* with the genus-1 collision residue, and of the $d\tau$-component with its $z$-derivative (Theorem 3.1).

(3) The identification of the genus-1 collision residue with the *Belavin–Drinfeld elliptic $r$-matrix* (Theorem 4.1).

(4) The *elliptic Gaudin commutativity*: the elliptic Gaudin Hamiltonians commute as a consequence of the classical elliptic Yang–Baxter equation (Theorem 5.1).

(5) The *class $M$ collision residue* for Virasoro and $\mathcal{W}_N$: higher-order elliptic operators that are genuinely new (Theorems 6.1 and 6.3).

(6) The *degeneration theorem*: all seven genus-1 faces degenerate continuously to their genus-0 counterparts as $q = e^{2\pi i\tau} \to 0$ (Theorem 7.1).

The paper is organized as follows. Section 2 introduces the genus-1 bar propagator and the elliptic collision residue. Section 3 proves the KZB identification (Face 4). Section 4 proves the elliptic $r$-matrix identification (Face 5). Section 5 proves the elliptic Gaudin commutativity (Face 7). Section 6 computes the class $M$ collision residue for Virasoro and $\mathcal{W}_N$. Section 7 establishes the degeneration to genus 0. Section 8 describes the computational verification.

# 2 The elliptic collision residue

## 2.1 The genus-1 bar propagator

At genus 0 the bar propagator is $\eta_{12} = d\log(z_1 - z_2)$: a meromorphic one-form on $\mathbb{P}^1$ with a simple pole along the diagonal. On the elliptic curve $E_\tau$, Liouville's theorem forbids a meromorphic function with a single simple pole that is doubly periodic. The propagator must be replaced by a quasi-periodic object.

**Definition 2.1** (Genus-1 bar propagator)**.** The *genus-1 bar propagator* on $E_\tau$ is the logarithmic derivative of the prime form:

$$\eta_{12}^{E_\tau} \;=\; d\log E(z_1,z_2) \;=\; \zeta_\tau(z_1 - z_2)\, d(z_1 - z_2), \tag{2.1}$$

where $E(z_1,z_2) = \theta_1(z_1 - z_2 \mid \tau)/\theta_1'(0 \mid \tau)$ is the prime form on $E_\tau$ (a section of $K^{-1/2} \boxtimes K^{-1/2}$), $\theta_1$ is the Jacobi theta function, and $\zeta_\tau$ is the Weierstrass zeta function for periods $(1,\tau)$:

$$\zeta_\tau(z) \;=\; \frac{1}{z} + \sum_{(m,n)\neq(0,0)} \left[\frac{1}{z - m - n\tau} + \frac{1}{m + n\tau} + \frac{z}{(m+n\tau)^2}\right]. \tag{2.2}$$

The Weierstrass zeta function has the Laurent expansion $\zeta_\tau(z) = 1/z + O(z^3)$ near $z = 0$, matching the genus-0 propagator $1/z$ at leading order. The difference is global: $\zeta_\tau$ is quasi-periodic,

$$\zeta_\tau(z+1) = \zeta_\tau(z) + 2\eta_1, \qquad \zeta_\tau(z+\tau) = \zeta_\tau(z) + 2\eta_\tau, \tag{2.3}$$

where $\eta_1 = \zeta_\tau(1/2)$ and $\eta_\tau = \zeta_\tau(\tau/2)$ are the quasi-periods, related by Legendre's identity $\eta_1\tau - \eta_\tau = \pi i$. The quasi-periodicity of the propagator is the geometric origin of the $B$-cycle monodromy that distinguishes genus 1 from genus 0 in all seven faces.

*Remark* 2.2 (Propagator weight). The bar propagator $d\log E(z,w)$ has weight 1 in both variables, regardless of the conformal weight of the fields being sewed. The logarithmic derivative of the prime form is a $(1,0)$-form in each variable; the conformal weight $h$ of the generator enters the OPE-mode coefficient $c_n$, not the propagator kernel.

## 2.2 The collision residue

**Definition 2.3** (Genus-1 collision residue)**.** For a chirally Koszul chiral algebra $\mathcal{A}$ on $E_\tau$, the *genus-*1 *collision residue* is

$$r^{(1)}_{\mathcal{A}}(z,\tau) \;:=\; \mathrm{Res}^{\mathrm{coll}}_{1,2}(\Theta_{\mathcal{A}})\big|_{E_\tau} \;\in\; \mathrm{End}_{\mathcal{A}}(2)[\![z^{-1}]\!] \otimes \mathbb{C}[\![\tau]\!], \tag{2.4}$$

where $\Theta_{\mathcal{A}}$ is the bar-intrinsic Maurer–Cartan element.

**Theorem 2.4** (Elliptic regularization)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra with genus-*0 *collision residue $r_{\mathcal{A}}(z) = \sum_{n\geq 0} c_n/z^{n+1}$, where $c_n = a_{(n)}b$ are the OPE-mode coefficients. The genus-*1 *collision residue is*

$$r^{(1)}_{\mathcal{A}}(z,\tau) \;=\; c_0\,\zeta_\tau(z) \;+\; \sum_{n\geq 1} \frac{(-1)^{n-1}}{n!}\, c_n \cdot \wp^{(n-1)}(z,\tau), \tag{2.5}$$

*where $\wp^{(m)} = d^m\wp/dz^m$. In particular, the first three terms are*

$$r^{(1)}_{\mathcal{A}}(z,\tau) \;=\; c_0\,\zeta_\tau(z) \;+\; c_1\,\wp(z,\tau) \;-\; \tfrac{1}{2}\,c_2\,\wp'(z,\tau) \;+\; \cdots \tag{2.6}$$

*The Laurent expansion of $r^{(1)}_{\mathcal{A}}$ near $z = 0$ agrees with $r_{\mathcal{A}}(z)$ at leading order.*

*Proof.* Each rational function $1/z^{n+1}$ $(n \geq 0)$ in the genus-0 collision expansion is the unique meromorphic function on $\mathbb{P}^1$ with a pole of order $n+1$ at $z = 0$ and vanishing at $z = \infty$. On $E_\tau$, the unique meromorphic function matching $1/z + O(z)$ at $z = 0$ with the prescribed quasi-periodicity is $\zeta_\tau(z)$ (for $n = 0$), while for $n \geq 1$ the unique doubly periodic function matching $1/z^{n+1} + O(z^{-n+1})$ is $(-1)^{n-1}\wp^{(n-1)}(z)/n!$. The prefactor is fixed by the Laurent expansion $\wp^{(m)}(z) = (-1)^m(m+1)!/z^{m+2} + O(1)$ at $m = n-1$:

$$\frac{(-1)^{n-1}}{n!}\,\wp^{(n-1)}(z) \;=\; \frac{(-1)^{n-1}}{n!}\cdot\frac{(-1)^{n-1}\,n!}{z^{n+1}} + O(z^{-n+1}) \;=\; \frac{1}{z^{n+1}} + O(z^{-n+1}).$$

Substituting into the propagator-extraction formula of Definition 2.1 yields (2.5). □

*Remark* 2.5 (Quasi-periodicity dichotomy). The $\wp$-function and all its derivatives are doubly periodic. The quasi-periodicity of $r^{(1)}_{\mathcal{A}}(z,\tau)$ is therefore inherited entirely from the $\zeta_\tau$-term:

$$r^{(1)}_{\mathcal{A}}(z+\tau,\tau) - r^{(1)}_{\mathcal{A}}(z,\tau) \;=\; 2\eta_\tau \cdot c_0. \tag{2.7}$$

When $c_0 = 0$ (the Cartan-type case: $\beta\gamma$, lattice VOAs), the collision residue is a genuine meromorphic function on $E_\tau$. When $c_0 \neq 0$ (the Yangian-type case: affine Kac–Moody, Virasoro, $\mathcal{W}_N$), it is quasi-periodic with non-trivial $B$-cycle monodromy.

## 3 Face 4: the KZB connection

At genus 0, Face 4 identifies the collision residue with the Gaiotto–Zeng commuting sphere Hamiltonians. At genus 1, the sphere Hamiltonians are replaced by the Knizhnik–Zamolodchikov–Bernard connection on the configuration space $\mathrm{Conf}_n(E_\tau)$. The KZB connection has two components: a $dz$-component generalizing the genus-0 KZ connection, and a $d\tau$-component encoding the dependence of conformal blocks on the modulus $\tau$.

**Theorem 3.1** (Face 4: KZB connection)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at level $k \neq -h^\vee$, the modular shadow connection on $\mathrm{Conf}_n(E_\tau)$ restricts to the KZB connection:*

$$\nabla^{\mathrm{KZB}} \;=\; d \;-\; \underbrace{\frac{1}{k+h^\vee}\sum_{i<j}\Omega_{ij}\,\zeta_\tau(z_{ij})\,dz_{ij}}_{dz\text{-component: genus-1 collision residue}} \;-\; \underbrace{\frac{1}{k+h^\vee}\sum_{i<j}\Omega_{ij}\,\wp(z_{ij},\tau)\,d\tau}_{d\tau\text{-component: modular Hamiltonian}}\;, \tag{3.1}$$

*where $z_{ij} = z_i - z_j$ and $\Omega_{ij}$ is the Casimir acting on the $i$-th and $j$-th tensor factors.*

(i) *The $dz$-component equals the genus-1 collision residue:*

$$H^{\mathrm{KZB}}_{z,i} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,\zeta_\tau(z_{ij}) \;=\; \sum_{j\neq i}\big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_{ij},\tau)\big]_{ij}. \tag{3.2}$$

(ii) *The $d\tau$-component equals the $z$-derivative of the collision residue, through the Weierstrass identity $\wp(z,\tau) = -\zeta'_\tau(z)$:*

$$H^{\mathrm{KZB}}_{\tau,i} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,\wp(z_{ij},\tau) \;=\; -\sum_{j\neq i}\partial_z\big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_{ij},\tau)\big]_{ij}. \tag{3.3}$$

*Proof.* **Part (i).** The affine Kac–Moody OPE $J^a(z)\,J^b(w) \sim k\,\delta^{ab}/(z-w)^2 + f^{ab}_{\;\;c}\,J^c(w)/(z-w)$ has $c_0 = \Omega/(k+h^\vee)$ and $c_n = 0$ for $n \geq 1$: the bar propagator absorbs one pole order, so the only surviving term is $c_0$. By Theorem 2.4, the elliptic regularization gives $r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) = \Omega\,\zeta_\tau(z)/(k+h^\vee)$. Substituting into the Gaiotto–Zeng formula produces (3.2), which is the $dz$-component of the KZB connection [3, 7].

**Part (ii).** The $d\tau$-component involves the modular Hamiltonian $D_i$, whose leading term is $\sum_{j\neq i}\wp(z_{ij},\tau)\,\Omega_{ij}/(k+h^\vee)$. The Weierstrass identity $\wp(z,\tau) = -\zeta'_\tau(z)$ identifies this with $-\partial_z r^{(1)}_{\mathcal{A}}$, yielding (3.3). □

*Remark* 3.2 (Shadow connection interpretation). The KZB connection is the genus-1 instance of the modular shadow connection $\nabla^{\mathrm{hol}}_{g,n}$ on $\overline{\mathcal{M}}_{g,n}$: at genus 0 it gives the KZ connection, at genus 1 the KZB connection, and at genus $\geq 2$ it produces new flat connections on configuration spaces of higher-genus curves.

## 4 Face 5: the elliptic $r$-matrix

At genus 0, Face 5 identifies the collision residue with the Drinfeld $r$-matrix $\Omega/((k+h^\vee)z)$. At genus 1, the rational $r$-matrix is replaced by the classical elliptic $r$-matrix of Belavin [1] and Belavin–Drinfeld [2].

**Theorem 4.1** (Face 5: elliptic $r$-matrix)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ with $\mathfrak{g}$ simple and $k \neq -h^\vee$, the genus-1 collision residue equals the classical elliptic $r$-matrix up to the standard level normalization:*

$$r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) \;=\; \frac{1}{k+h^\vee}\, r^{\mathrm{ell}}_{\mathfrak{g}}(z,\tau). \tag{4.1}$$

*For $\mathfrak{g} = \mathfrak{sl}_2$, the Belavin $r$-matrix is*

$$r^{\mathrm{ell}}_{\mathfrak{sl}_2}(z,\tau) \;=\; \zeta_\tau(z)\cdot\frac{H\otimes H}{2} \;+\; \phi_+(z,\tau)\,E\otimes F \;+\; \phi_-(z,\tau)\,F\otimes E, \tag{4.2}$$

*where $\{H,E,F\}$ is the standard basis of $\mathfrak{sl}_2$ and the theta-function ratios are*

$$\phi_\pm(z,\tau) \;=\; \frac{\theta_1'(0\mid\tau)\,\theta_1(z\pm\frac{1}{2}\mid\tau)}{\theta_1(z\mid\tau)\,\theta_1(\pm\frac{1}{2}\mid\tau)}. \tag{4.3}$$

*The elliptic $r$-matrix satisfies the classical Yang–Baxter equation on $E_\tau$:*

$$[r^{\mathrm{ell}}_{12}(z_1-z_2),\, r^{\mathrm{ell}}_{13}(z_1-z_3)] + [r^{\mathrm{ell}}_{12}(z_1-z_2),\, r^{\mathrm{ell}}_{23}(z_2-z_3)] + [r^{\mathrm{ell}}_{13}(z_1-z_3),\, r^{\mathrm{ell}}_{23}(z_2-z_3)] = 0. \tag{4.4}$$

*Proof.* For the Cartan channel: the Kac–Moody OPE with $c_0 = \Omega/(k+h^\vee)$ gives, after genus-1 regularization, $\zeta_\tau(z)\cdot H\otimes H/2$ in the Cartan direction.

For the root channels: on $E_\tau$, the chiral fields $J^E$, $J^F$ associated to the positive and negative roots satisfy twisted boundary conditions under $B$-cycle monodromy. The collision residue in the root channels acquires the theta-function ratios $\phi_\pm(z,\tau)$ (4.3), which are the unique meromorphic functions on $\mathbb{C}$ with a simple pole at $z=0$ matching $1/z$ and with $B$-cycle quasi-periodicity dictated by the root lattice [1].

The classical Yang–Baxter equation (4.4) follows from the collision-depth-2 projection of the Maurer–Cartan equation at genus 1: the infinitesimal braid relation $[\Omega_{ij}, \Omega_{ik}+\Omega_{jk}] = 0$ is replaced by its elliptic counterpart. The overall identification (4.1) follows from Theorem 2.4 combined with uniqueness of the classical elliptic $r$-matrix [2]. □

*Remark* 4.2 (Higher rank). For $\mathfrak{g} = \mathfrak{sl}_N$ with $N \geq 3$, the Belavin elliptic $r$-matrix is a matrix-valued meromorphic function on $E_\tau$ with poles at $z=0$ whose residue is the $\mathfrak{sl}_N$ Casimir. The collision residue produces this $r$-matrix by the same mechanism: elliptic regularization of the Cartan and root channels, with theta-function ratios $\phi_\alpha(z,\tau)$ indexed by the roots $\alpha$. The identification (4.1) holds for all simple $\mathfrak{g}$.

## 5 Face 7: the elliptic Gaudin model

At genus 0, Face 7 identifies the collision residue with the Gaudin Hamiltonians $H_i^{\mathrm{Gaudin}} = \sum_{j\neq i}\Omega_{ij}/(z_i-z_j)$. At genus 1, the Gaudin model acquires elliptic structure.

**Theorem 5.1** (Face 7: elliptic Gaudin Hamiltonians)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$, the elliptic Gaudin Hamiltonians on* $\mathrm{Conf}_n(E_\tau)$ *are*

$$H_i^{\mathrm{ell}} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,\zeta_\tau(z_i-z_j) \;=\; \sum_{j\neq i}\big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_i-z_j,\tau)\big]_{ij}. \tag{5.1}$$

*They mutually commute:* $[H_i^{\mathrm{ell}}, H_j^{\mathrm{ell}}] = 0$ *for all* $i, j$.

*Proof.* The identification of $H_i^{\mathrm{ell}}$ with the collision residue is immediate from Theorem 2.4: $r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) = \Omega\,\zeta_\tau(z)/(k+h^\vee)$.

For commutativity, the Maurer–Cartan equation at genus 1 gives $D_{\mathcal{A}}\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}},\Theta_{\mathcal{A}}] = 0$ restricted to the collision-depth-2 stratum of $\overline{\mathcal{M}}_{1,n}$. Expanding the bracket and projecting onto the $(i,j)$ and $(i,k)$ channels produces the elliptic infinitesimal braid relation

$$[\zeta_\tau(z_{ij})\,\Omega_{ij},\;\; \zeta_\tau(z_{ik})\,\Omega_{ik} + \zeta_\tau(z_{jk})\,\Omega_{jk}] \;=\; 0, \tag{5.2}$$

which follows from the classical elliptic YBE (4.4) by specializing to three points and evaluating on the Casimir. The commutativity $[H_i^{\mathrm{ell}}, H_j^{\mathrm{ell}}] = 0$ is the $n$-point consequence of (5.2). □

*Remark* 5.2 (Hitchin system). The elliptic Gaudin model is the genus-1 instance of the Hitchin integrable system on $\mathrm{Bun}_G(E_\tau)$. The Hitchin Hamiltonians are sections of a line bundle on the base of the Hitchin fibration; at genus 1, the base is the affine space of elliptic spectral curves, and the Hitchin Hamiltonians restrict to the elliptic Gaudin Hamiltonians on the trivial $G$-bundle. At genus $g \geq 2$, the collision residue on $\Sigma_g$ produces the Hitchin Hamiltonians on $\mathrm{Bun}_G(\Sigma_g)$; this is the structural target of the holographic programme [8].

# 6 Class $M$ at genus 1: higher-order elliptic operators

For affine Kac–Moody algebras (class $L$), the collision expansion terminates at $c_0$: the collision depth is $k_{\max} = 1$, and only the $\zeta_\tau$-term contributes. For class $M$ algebras (Virasoro, $\mathcal{W}_N$), the OPE has poles of higher order, and the collision expansion involves $\wp$, $\wp'$, and higher derivatives of the Weierstrass functions.

## 6.1 Virasoro at genus 1

The Virasoro algebra has maximal OPE pole order $p_{\max} = 4$ (from the $T(z)\,T(w)$ OPE). The bar propagator absorbs one order, giving collision depth $k_{\max} = 3$. The genus-0 collision residue is

$$r_{\mathrm{Vir}_c}(z) \;=\; \frac{c/2}{z^3} \;+\; \frac{2T}{z}, \tag{6.1}$$

with $c_0 = 2T$, $c_1 = 0$ (the $z^{-2}$ pole is absent from the $r$-matrix by a parity argument: the $d\log$ extraction sends $z^{-2n}$ to $z^{-(2n-1)}$), and $c_2 = c/2$.

**Theorem 6.1** (Virasoro genus-1 collision residue)**.** *The genus-1 collision residue of* $\mathrm{Vir}_c$ *is*

$$r^{(1)}_{\mathrm{Vir}_c}(z,\tau) \;=\; 2T\cdot\zeta_\tau(z) \;-\; \frac{c}{4}\,\wp'(z,\tau). \tag{6.2}$$

*This is an elliptic differential operator of order 2 on conformal blocks on $E_\tau$: the $\zeta_\tau$-term acts by multiplication, while the $\wp'$-term acts as a second-order differential operator through the stress-energy Ward identity.*

*Proof.* The genus-0 collision expansion has $c_0 = 2T$, $c_1 = 0$, $c_2 = c/2$. By Theorem 2.4:

$$r^{(1)}_{\mathrm{Vir}_c}(z,\tau) \;=\; c_0\,\zeta_\tau(z) + c_1\,\wp(z,\tau) - \tfrac{1}{2}\,c_2\,\wp'(z,\tau).$$

Since $c_1 = 0$, the $\wp$-term vanishes, and the result is (6.2). The Laurent expansion near $z = 0$ confirms consistency: $2T/z - (c/4)\cdot(-2/z^3 + O(z)) = (c/2)/z^3 + 2T/z + O(1)$, matching $r_{\mathrm{Vir}_c}(z)$ at leading order.

The $\wp'$-term acts as a second-order differential operator because the Virasoro OPE at order $(z-w)^{-4}$ involves $(c/2)\cdot\mathrm{id}$, which through the Ward identity produces a second derivative $\partial_z^2$ on the correlator; the elliptic regularization replaces $1/z^4$ by $-\wp'(z,\tau)/2$. □

*Remark* 6.2 (The $c_1 = 0$ vanishing). The vanishing of $c_1$ is a parity phenomenon: the OPE mode $T_{(2)}T = \partial T$ is a total derivative, and its contribution to the $r$-matrix vanishes after $d\log$-extraction. In the elliptic regularization, this means the $\wp(z,\tau)$-term is absent. The first non-trivial correction beyond $\zeta_\tau$ involves $\wp'$, not $\wp$.

### 6.2 $\mathcal{W}_N$ at genus 1

The $\mathcal{W}_N$ algebra has generators $W_2 = T, W_3, \ldots, W_N$ of conformal weights $2, 3, \ldots, N$. The maximal OPE pole order among generating fields is $p_{\max} = 2N$ (from the $W_N$-$W_N$ OPE), giving collision depth $k_{\max} = 2N - 1$.

**Theorem 6.3** ($\mathcal{W}_N$ genus-1 collision residue)**.** *The genus-1 collision residue of $\mathcal{W}_N$ involves the Weierstrass functions up to $\wp^{(2N-3)}$:*

$$r^{(1)}_{\mathcal{W}_N}(z,\tau) \;=\; c_0\,\zeta_\tau(z) \;+\; \sum_{\substack{n=1\\ c_n\neq 0}}^{2N-2} \frac{(-1)^{n-1}}{n!}\,c_n\cdot\wp^{(n-1)}(z,\tau), \tag{6.3}$$

*where $c_n$ are the OPE-mode collision coefficients. The highest-order term is an elliptic differential operator of order $2N-2$. For $\mathcal{W}_3$: the collision expansion involves $\zeta_\tau, \wp, \wp', \wp'', \wp'''$, producing fourth-order elliptic differential operators.*

*Proof.* Direct specialization of Theorem 2.4 to the $\mathcal{W}_N$ OPE data with $p_{\max} = 2N$: the collision depth $k_{\max} = 2N-1$ means the genus-0 expansion $r_{\mathcal{W}_N}(z) = \sum_{n=0}^{2N-2} c_n/z^{n+1}$ has $2N-1$ terms, each regularized by $1/z \mapsto \zeta_\tau(z)$ and $1/z^{n+1} \mapsto (-1)^{n-1}\wp^{(n-1)}(z)/n!$ for $n \geq 1$. □

*Remark* 6.4 (Genuinely new operators). The elliptic differential operators produced by the class $M$ collision residue are not present in the existing integrable systems literature. The Hitchin system on $\mathrm{Bun}_G(E_\tau)$ involves only the $\zeta_\tau$-term (collision depth 1, corresponding to the Casimir of $\mathfrak{g}$). The KZB connection involves $\zeta_\tau$ and $\wp$ (depths 1 and 2). The class $M$ collision residue at depth $\geq 3$ produces operators involving $\wp', \wp'', \ldots, \wp^{(2N-3)}$ that arise from higher OPE poles specific to $\mathcal{W}$-algebras and have no counterpart in the theory of affine Kac–Moody algebras.

## 7 Degeneration to genus 0

**Theorem 7.1** (Degeneration)**.** *As $\mathrm{Im}(\tau) \to \infty$ (equivalently, $q = e^{2\pi i\tau} \to 0$), all seven genus-1 faces degenerate to their genus-0 counterparts:*

(i) (Propagator.) $\zeta_\tau(z) \to \pi\cot(\pi z) \to 1/z + O(z)$.

(ii) (Weierstrass functions.) $\wp(z,\tau) \to \pi^2/\sin^2(\pi z) \to 1/z^2 + O(1)$, *and* $\wp^{(m)}(z,\tau) \to (-1)^m(m+1)!/z^{m+2} + O(z^{-m})$.

(iii) (KZB $\to$ KZ.) *The $dz$-component degenerates to the KZ connection: $\zeta_\tau(z_{ij}) \to 1/z_{ij}$. The $d\tau$-component has no genus-0 counterpart.*

(iv) (Elliptic $\to$ rational $r$-matrix.) $r^{\mathrm{ell}}_{\mathfrak{g}}(z,\tau)/(k+h^\vee) \to k\,\Omega/((k+h^\vee)z) = \Omega/((k+h^\vee)z)$ *through the intermediate trigonometric $r$-matrix: $r^{\mathrm{ell}} \to r^{\mathrm{trig}} \to r^{\mathrm{rat}}$.*

(v) (Elliptic $\to$ rational Gaudin.) $H_i^{\mathrm{ell}} \to H_i^{\mathrm{Gaudin}} = \sum_{j\neq i}\Omega_{ij}/(z_i - z_j)$.

*Proof.* As $q \to 0$, the Weierstrass functions have the standard degenerations. The zeta function: $\zeta_\tau(z) = \pi\cot(\pi z) + O(q)$, giving $1/z + O(z)$ for $|z| < 1$. The $\wp$-function: $\wp(z,\tau) = \pi^2/\sin^2(\pi z) - 2\pi^2 E_2(\tau)/3 + O(q)$, giving $1/z^2 + O(1)$. Higher derivatives degenerate analogously.

The KZB degeneration: with $\zeta_\tau(z) \to 1/z$, the $dz$-component (3.2) becomes $\sum_{j\neq i}\Omega_{ij}/((k+h^\vee)z_{ij})\,dz_{ij}$, which is the KZ connection.

The elliptic $r$-matrix degeneration: as $q \to 0$, $\phi_\pm(z,\tau) \to \pi/\sin(\pi z) + O(q)$, recovering the trigonometric $r$-matrix. A further limit $z \mapsto z/L$, $L \to \infty$ gives $\pi/\sin(\pi z/L) \to L/z$, recovering the rational $r$-matrix $\Omega/((k+h^\vee)z)$ in the KZ normalization. Both limits are continuous in the topology of formal Laurent series. □

The degeneration theorem provides the consistency check between genus-0 and genus-1: every genus-1 face is a continuous deformation of the corresponding genus-0 face, with deformation parameter $q$. At $q = 0$, the elliptic curve degenerates to the nodal rational curve, and all seven identifications collapse to their genus-0 ancestors.

## 8 Computational verification

The genus-1 seven-face identifications are verified by the compute engine `theorem_genus1_seven_faces_engine.py` which implements the following tests.

(1) **Elliptic regularization.** For each standard family $\mathcal{A}$ in the landscape census, verify that the genus-1 collision residue (2.5) matches the elliptic regularization of the genus-0 collision expansion at $O(q^0)$ to machine precision.

(2) **KZB identification.** For $\mathcal{A} = \widehat{\mathfrak{sl}}_2$ at levels $k = 1, 2, 3, 5, 10$, verify that the $dz$-component of the KZB connection equals the collision residue, and that the $d\tau$-component equals $-\partial_z r^{(1)}_{\mathcal{A}}$.

(3) **Elliptic CYBE.** Verify the classical elliptic Yang–Baxter equation (4.4) numerically for $\mathfrak{sl}_2$ at 20 random evaluation points $(z_1, z_2, z_3) \in E_\tau^3$ and 5 values of $\tau$.

(4) **Gaudin commutativity.** Verify $[H_i^{\mathrm{ell}}, H_j^{\mathrm{ell}}] = 0$ for $\mathfrak{sl}_2$ on $\mathrm{Conf}_3(E_\tau)$ at 10 random configurations and 5 values of $\tau$.

(5) **Degeneration.** Verify that $\|r^{(1)}_{\mathcal{A}}(z,\tau) - r_{\mathcal{A}}(z)\|/\|r_{\mathcal{A}}(z)\| < 10^{-6}$ for $\mathrm{Im}(\tau) \geq 12$ across all standard families.

(6) **Class $M$ structure.** For $\mathrm{Vir}_c$ at $c = 1, 1/2, 25, 26$, verify (6.2) and confirm the $c_1 = 0$ vanishing. For $\mathcal{W}_3$ at $c = 2, 50, 98$, verify (6.3) and confirm terms through $\wp'''$.

# 9 The Verlinde formula from ordered chiral homology

At integer level $k \in \mathbb{Z}_{\geq 1}$, the ordered chiral homology of $\widehat{\mathfrak{g}}_k$ on $E_\tau$ recovers the Verlinde partition function. The connection to the seven-face programme is direct: the KZB connection of Theorem 3.1 has finite monodromy at integer level, and the trace of the $B$-cycle monodromy over the space of conformal blocks recovers the Verlinde formula.

**Proposition 9.1** (Verlinde from genus-1 collision residue)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at integer level $k \geq 1$, the $B$-cycle monodromy of the KZB connection is*

$$\mathcal{M}_B \;=\; \exp\Big(\frac{2\eta_\tau}{k+h^\vee}\,\Omega\Big) \;=\; \exp\big(2\eta_\tau \cdot c_0\big), \tag{9.1}$$

*where $c_0 = \Omega/(k+h^\vee)$ is the degree-0 collision coefficient. The quantum group parameter is $q = \exp(2\pi i/(k+h^\vee))$, and the Verlinde partition function at genus 1 is*

$$Z_1(\widehat{\mathfrak{g}}_k) \;=\; \sum_{\lambda \in P_+^k} |S_{0\lambda}|^0 \;=\; |P_+^k|, \tag{9.2}$$

*where $P_+^k$ is the set of integrable highest weights at level $k$ and $S_{0\lambda}$ are the modular $S$-matrix entries. At higher genus $g$, the Verlinde formula $Z_g = \sum_\lambda S_{0\lambda}^{2-2g}$ is recovered from the ordered chiral homology via handle attachment and separating factorization of the ordered bar complex.*

*Remark* 9.2 (The $B$-cycle monodromy and the quantum group). The quasi-periodicity (2.7) of the genus-1 collision residue is the geometric origin of the quantum group parameter. For affine $\widehat{\mathfrak{sl}}_{2,k}$: the $B$-cycle monodromy eigenvalues are $q^{j(j+1)}$ for $j = 0, 1/2, \ldots, k/2$, where $q = e^{2\pi i/(k+2)}$ is a root of unity. The passage from the $E_1$ collision residue $r^{(1)}(z,\tau)$ to the quantum group $U_q(\mathfrak{g})$ is the $B$-cycle monodromy of the ordered bar complex: the spectral parameter $z$ (genus-0 data) becomes the quantum group parameter $q$ (genus-1 data) through exponentiation of the quasi-period.

# 10 The critical level

The seven-face programme at genus 1 has a distinguished boundary: the critical level $k = -h^\vee$, where the Sugawara construction degenerates. The collision residue and the KZB connection undergo a qualitative change at this value.

**Theorem 10.1** (Critical level jump at genus 1)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at the critical level $k = -h^\vee$:*

(i) *The shadow invariant vanishes: $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$. All monodromy eigenvalues of the genus-1 collision residue become integers.*

(ii) *The first cohomology of the ordered chiral homology doubles: $\dim H^1$ jumps from 4 (generic level) to 8 (critical level) for $\mathfrak{g} = \mathfrak{sl}_2$.*

(iii) *The bar cohomology at critical level computes the algebra of functions on the space of* $\mathfrak{g}$*-opers:* $H^*(B(\widehat{\mathfrak{g}}_{-h^\vee})) \cong \Omega^*(\mathrm{Op}_{\mathfrak{g}}(D))$.

(iv) *The genus-*1 *collision residue degenerates: the KZ normalization* $r^{(1)}(z,\tau) = \Omega\,\zeta_\tau(z)/(k+h^\vee)$ *has a pole at* $k=-h^\vee$*, and the KZB connection becomes singular. The critical level requires a separate regularization via the Feigin–Frenkel center [5].*

*Remark* 10.2 (Koszulness at critical level). At generic level $k \neq -h^\vee$, the affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ is chirally Koszul, and the seven-face programme applies without modification. At the critical level $k=-h^\vee$, chirally Koszulness fails: the center of $\widehat{\mathfrak{g}}_{-h^\vee}$ jumps (it becomes the Feigin–Frenkel center, isomorphic to the algebra of functions on the space of opers), and the bar complex acquires additional cohomology. The seven faces degenerate: the Gaudin Hamiltonians become the Feigin–Frenkel higher Gaudin Hamiltonians, and the $r$-matrix limit requires the Langlands dual group ${}^LG$ via the geometric Langlands correspondence. The critical level is not an obstruction to the programme; it is a boundary where new structure emerges.

*Remark* 10.3 (Spectral sequence at critical level). The passage from generic to critical level is controlled by a spectral sequence with $d_1 = \lambda \cdot [\delta]$, where $\lambda = k + h^\vee$ is the level shift and $\delta$ is the boundary class. At $\lambda = 0$ (critical level), $d_1$ vanishes, and the spectral sequence degenerates at $E_1$: the bar cohomology jumps because the differential that would kill the extra classes is absent. This is the mechanism by which $H^1$ doubles from 4 to 8 for $\mathfrak{sl}_2$.

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